

Garlic

CULTIVARS

- Music
- Siberian

INTENSIVE SPACING

- 3 rows (5" - 12 cm) per bed spaced every 5.5" (14 cm) on the row.

FERTILIZATION

- Garlic is a crop worth amending with lots of compost, +/-1 inch (5 cm) all across the bed. When planting garlic in the fall, it's important to protect the soil with straw in order to avoid fertility leaching over the winter. Alternatively mulch with hay mulch (without seeds) which supplies nitrogen to garlic at double the rate of straw (but it breaks down much faster and should be applied in larger amounts).
- Come springtime, add a foliar application of a rich organic fertilizer like fish emulsion (5-1-1) to help the plant quickly develop its canopy.

IRRIGATION

Irrigation Type

- Drip tape (4 per bed) or sprinkler

Irrigation Management

- Root depth: Shallow (6"-24")
- General: Likes steady supply of water. Stop watering several weeks before harvest to reduce succulence and therefore reduce rot during drying. Critical to water during bulb enlargement.

IMPLANTATION

We grow fall garlic and plant from the first fall frost date until as late as November. The objective is that the plant develops good roots, but no top growth before freezing temperatures set in. Compared to spring planting, this method produces larger bulb size.

EQUIPMENT

- Rotary harrow
- Broadfork
- 30" bed prep rake and tubes
- Garlic dibbler
- Weed free straw, 4 bales per 100' (30M) beds

SEEDING ITINERARY

- First, separate cloves from the bulb. This should be done the day prior or the same day of the planting. Make sure to discard anything that is too small, rotten or black. The largest cloves will make the largest bulbs.
- Prepare seed beds by first passing through with the broadfork.
- Uniformly apply compost and pass with the rotary harrow at a depth of 2" (5 cm). To make seeding the garlic cloves easier (seeing as the soil is loose after being harrowed), we only prepare the beds that will be seeded that day.
- Mark the 3 rows using the bed prep rake and the spacings on the rows using the garlic planter. Make sure that all holes are +/- 2.5 " (6 cm) deep.
- 2 people work face to face planting the cloves while moving up the aisle. Push the clove, root end down, all the way into the planting holes. The tops of the cloves are pointing up.
- Rake over with the back of the rake and cover the beds of at least 6" (15 cm) of straw mulch.

WEEDING

- Keep the growing area free of weeds by hand weeding once or twice. Garlic does not compete well with weeds, hence the importance of putting down enough straw to inhibit weed growth in the first place.
- Make sure to always use straw (or hay) that is weed free. First cut rye (which grows before all other cereals in northern climates) is a good choice.

PLANT PROTECTION

- Garlic is in the onion family and can have the same insect pests and diseases as onions. To minimize disease, practice a 3-5 year crop rotation out of Alliums.
- To fend off leek moths, Tricho-Guard cards should be placed in the gardens at a density of 4 per plot and changed weekly from mid-May to mid-June. These cards contain parasitoid wasp eggs which, when adult, lay inside the leek moth eggs.
- The quality of the seed stock also plays a big role in the quality. NEVER plant garlic cloves that might have been infected by fungal diseases.

HARVEST

- To harvest the scapes, wear tree planting bags and using both hands, snap them off the stem just above the main stalk of the garlic. Scapes should be longer than 6" (15 cm) and curled.
- Garlic is ready to be harvested when 3 to 4 leaves start to die off (yellow and dry). It's VERY important to harvest on a sunny day to prevent moisture-related problems when curing.
- Pull garlic out of the ground by grabbing the stalk as low as possible near the ground, brushing off the soil and making bundles.
- While some people are pulling, others will process the garlic in the fields. Ideally standing in a shaded area.
- Cut off the roots with a sharp cutter as close as possible below the bottom of the bulb. Also, cut the stem to about 2" (5 cm).

- Before the day ends, the harvested garlic is brought inside to be cured on a seeding table or piled in harvest crates. In both cases, make sure to allow for the best possible airflow.
- 1 person should harvest 800 garlic scape and 35 lbs garlic per person per hour.

CONDITIONING

- When curing the garlic in a greenhouse, make sure to cover it with shade cloth to protect garlic from sun scald. Garlic is laid flat on greenhouse benches with lots of fans blowing at all time in different directions.
- When curing the garlic in a heated room, make sure to control humidity by adding a dehumidifier and lots of fans blowing at all time in different directions. Garlic can be cured in stacked harvest containers given that containers are not over filled.
- Curing is complete when the outer skins are dry and crispy, the neck is constricted, and the center of the cut stem is hard.
- Cleaning Procedures; when the garlic is cured, peel it softly, removing one layer of paper (the papery wrapping helps the garlic store longer). Garlic is sold by unit or by weight, packaged in mesh bags with cloves of different size.
- Before selling the stock, make sure to select your own seeds first, putting aside the biggest bulbs in the desired quantity for next year's planting. Estimate +/- 13 lbs of seeds per bed.

STORING

- Garlic is stored in a cold room keep between 38 °F-50°F (3 °C-8°C) with about 65-70% relative humidity.